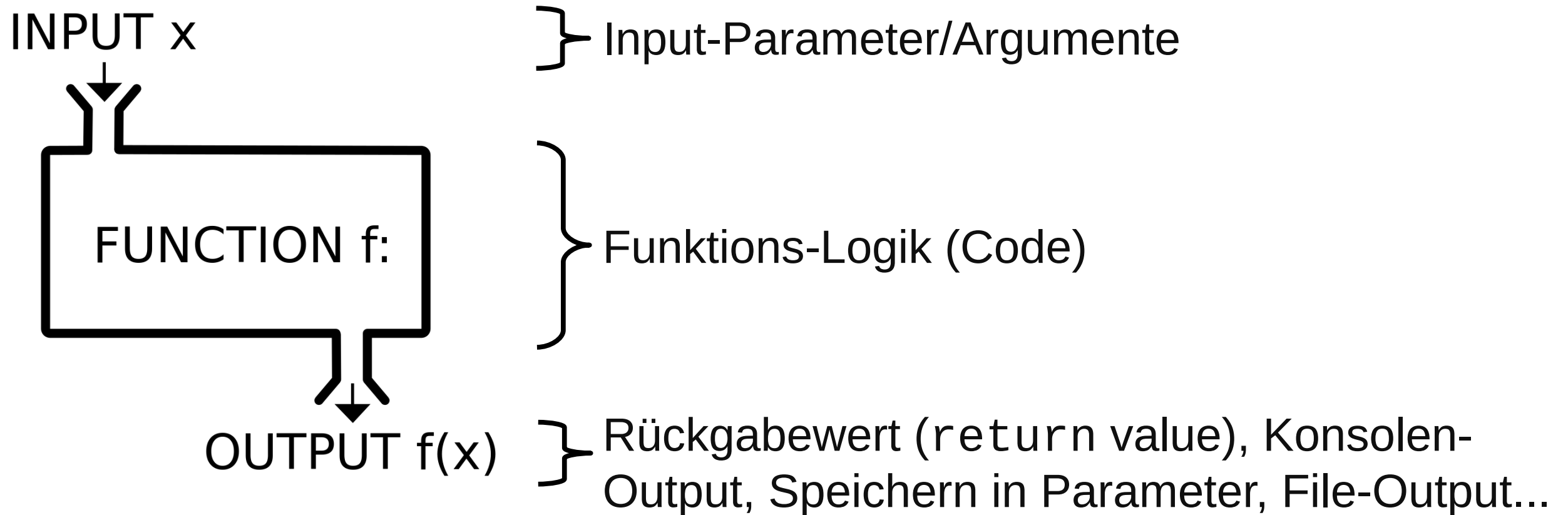


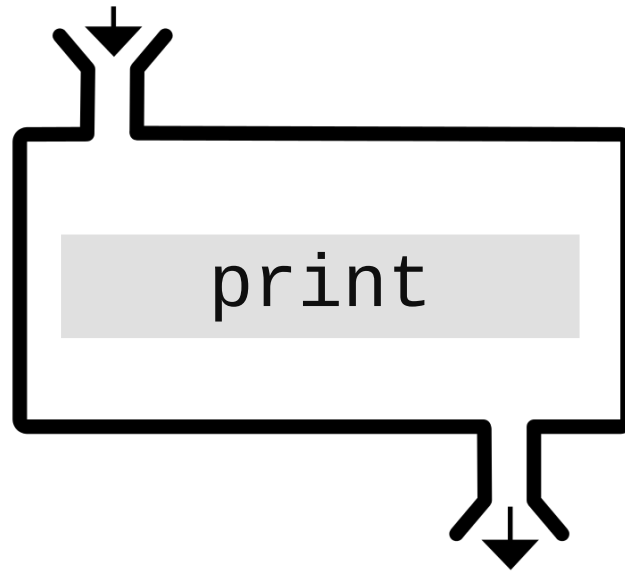
Functions (0): Grundkonzept



Functions (1):

Beispiel `print('This', 'is', 'a', 'Test')`

Argumente: `'This', 'is', 'a', 'Test'`

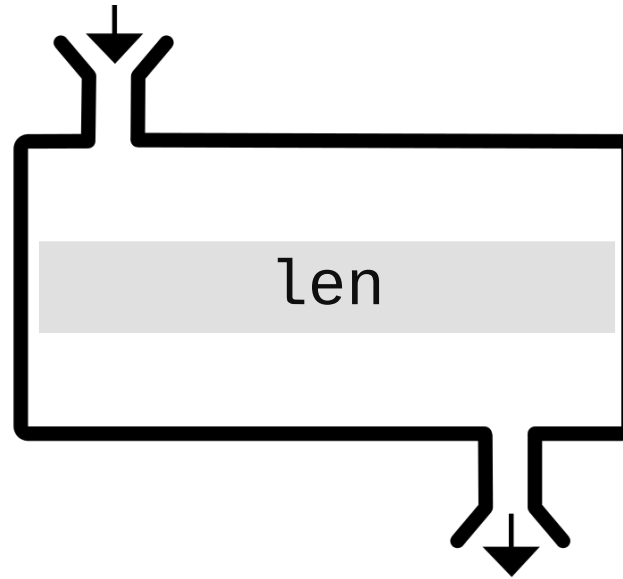


In Konsole: `This is a Test`

Functions (2):

Beispiel `len([4, 8, 'This', 'Test'])`

Argumente: `[4, 8, 'This', 'Test']`



Rückgabewert
(return-value)

4

Funktionen/Functions (3)

- „Benannter Codeblock“ mit spezifischer Aufgabe
 - Wiederverwendbar, einfacher lesbar, besser wartbar
 - Beispiele von Python-Funktionen: `len()`, `print()`, `input()`, ...
- Definiert durch Schlüsselwort **def**, Name der Funktion, Parameter-Liste
 - Code-Block in Funktion: eingerückt!
- Ausführung bei Aufruf (call), danach Programm fortgesetzt
- Rückgabe von Werten an Hauptprogramm möglich (`return`)

```
def test_function(param):  
    print(f'You passed "{param}" to this function')  
  
test_function(13)
```

Functions (4):

Einfaches Code-Beispiel für Definition und Aufruf

```
def test_simple():  
    print(f'Hi everyone! ')
```



```
def test_param(param):  
    print(f'You passed "{param}" to this function')
```



```
test_simple()  
test_param('Hello!')
```

Functions (7): Beispiele für valide Aufrufe

```
def say_name(first, last, country='Austria'):  
    print(f'Firstname is {first}, surname is {last}.')  
    print(f'I live in {country}.')
```

Passing arguments by position

```
say_name('Max', 'Muster')
```

```
say_name('Maria', 'Muster', 'Germany')
```

Passing arguments using keyword specification

```
say_name(first='Max', last='Muster')
```

```
say_name(country='Germany', first='Maria', last='Muster')
```

Mix of both

```
say_name('Maria', country='Germany', last='Muster')
```

Functions (9): Argument Errors (Beispiele)

```
def say_name(first, last, country='Austria'):
    print(f'Firstname is {first}, surname is {last}.')
    print(f'I live in {country}.')
```

```
say_name('Max')
```

```
# TypeError: say_name() missing 1 required positional argument: 'last_name'
```

```
say_name('Austria', 'Muster', 'Max')
```

```
# Kein Fehler in Ausführung, aber: "Logischer Fehler"
```

```
say_name(fs='Max', last='Muster')
```

```
# TypeError: say_name() got an unexpected keyword argument 'fs'
```

```
say_name(first='Max', 'Muster')
```

```
# SyntaxError: positional argument follows keyword argument
```

```
# There are many more possibilities for errors :-)
```